



Introduction to Google Colab



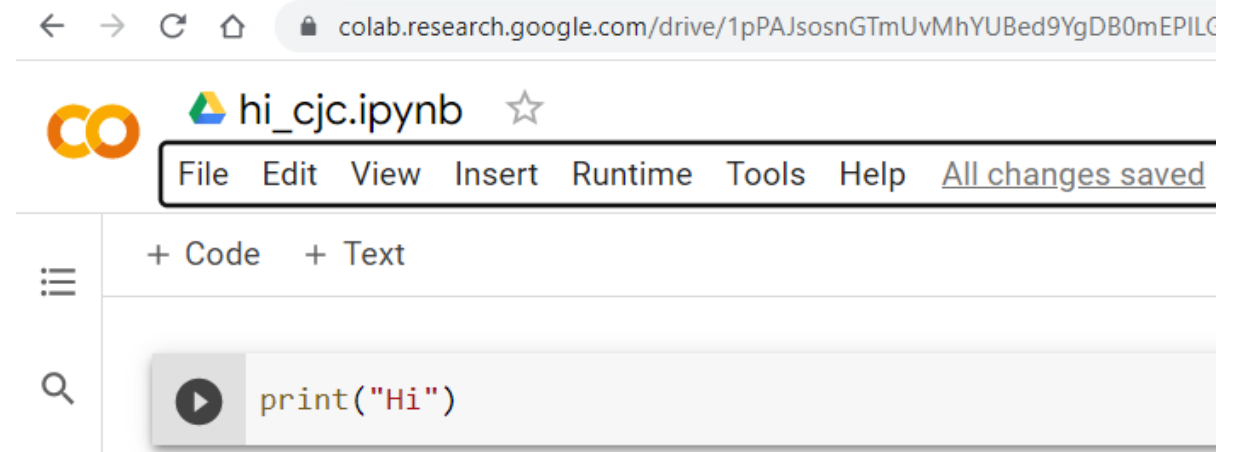
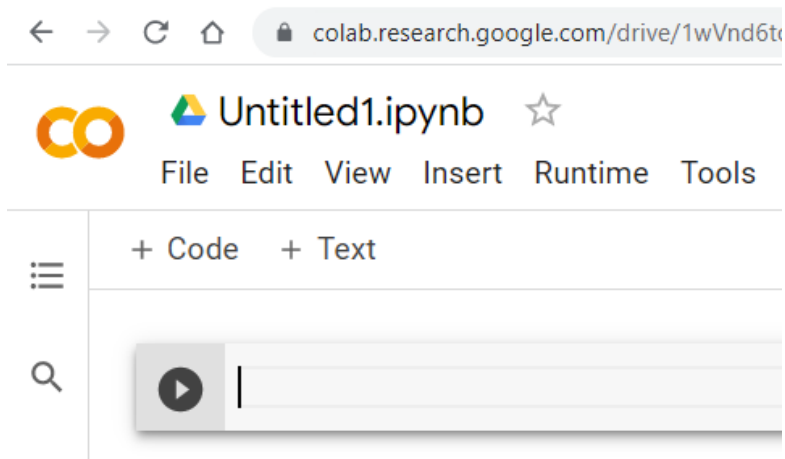


- Colaboratory is a free cloud research tool for machine learning with **free GPU (or TPU)**
- For Python Only
- It provides Jupyter notebook environment that requires no setup to use
- Develop deep learning applications with Python using popular libraries such as Keras, TensorFlow, PyTorch, and OpenCV

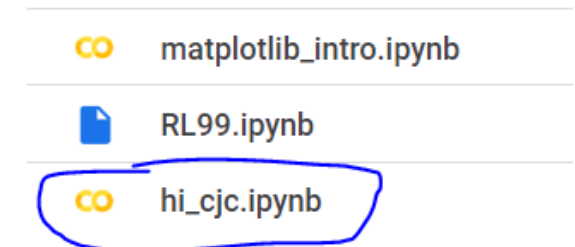
Getting started with Colab

1. Login to your Google account (LTU account is a Google account)
2. Go to: <https://colab.research.google.com>
3. Click on "New notebook" →

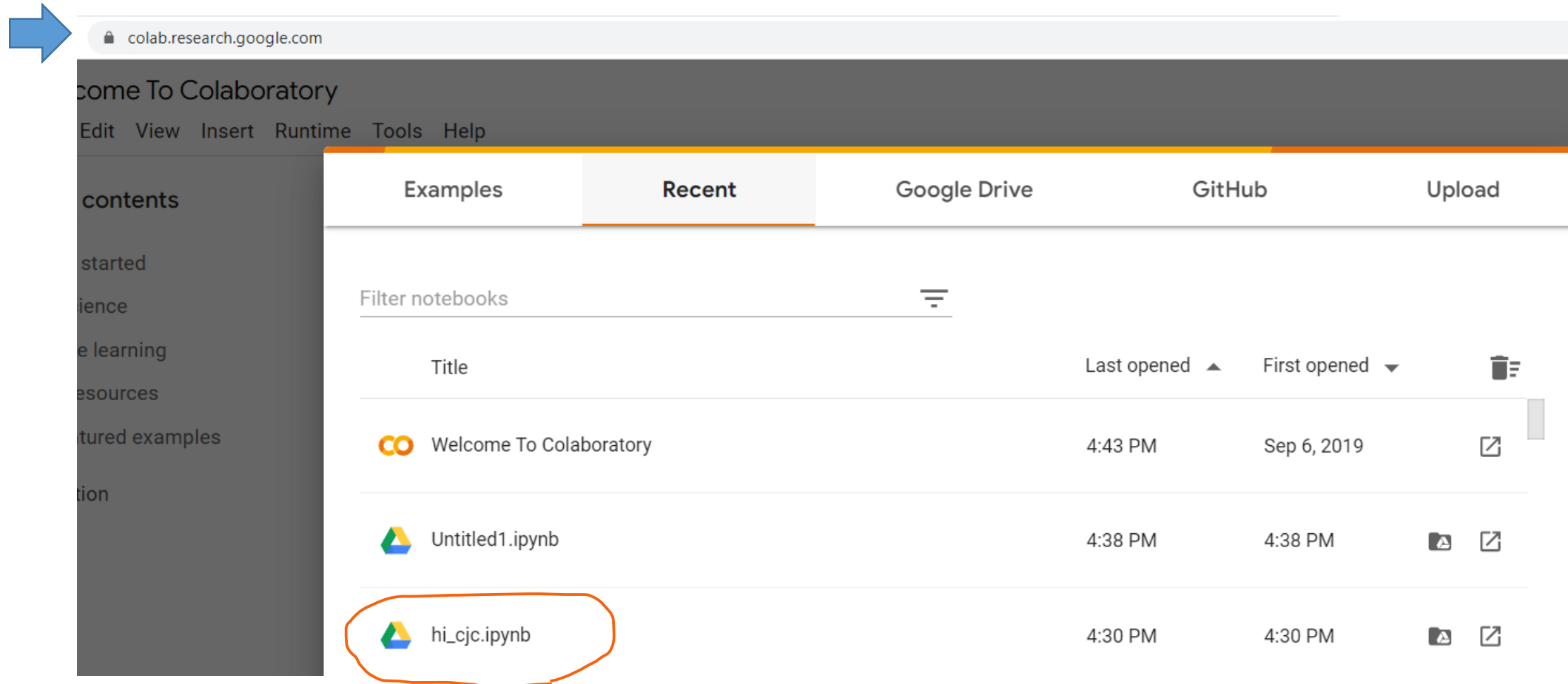
New notebook Cancel



Ctrl + s: file saved in My Drive/Colab Notebooks/HiWorld.ipynb



How to open saved ipynb files?



The screenshot shows the Google Colaboratory web interface. A blue arrow points to the address bar which contains 'colab.research.google.com'. The interface has a dark sidebar on the left with a 'contents' panel. The main area has a top navigation bar with 'Examples', 'Recent', 'Google Drive', 'GitHub', and 'Upload' tabs. The 'Recent' tab is active, showing a table of notebooks. The notebook 'hi_cjc.ipynb' is circled in orange.

Title	Last opened ▲	First opened ▼	
 Welcome To Colaboratory	4:43 PM	Sep 6, 2019	
 Untitled1.ipynb	4:38 PM	4:38 PM	 
 hi_cjc.ipynb	4:30 PM	4:30 PM	 



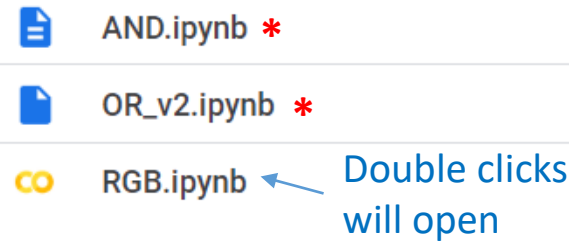
Watch: Google Colab Introduction. Colab Tutorial. Colab for Beginners. Colab Explained by AppliTorial (12 min)

<https://youtu.be/g0xu9DA4gDw>

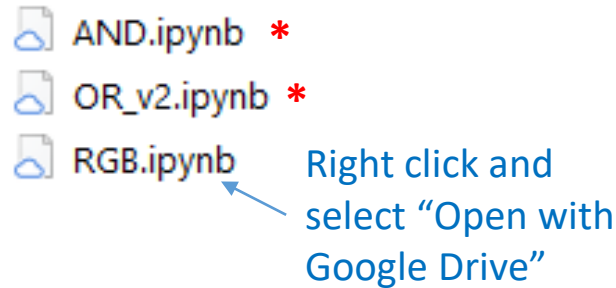
Learn how to mount your “My Drive”

To Open Colab *.ipynb files

Google Drive View



Google Drive for Desktop View

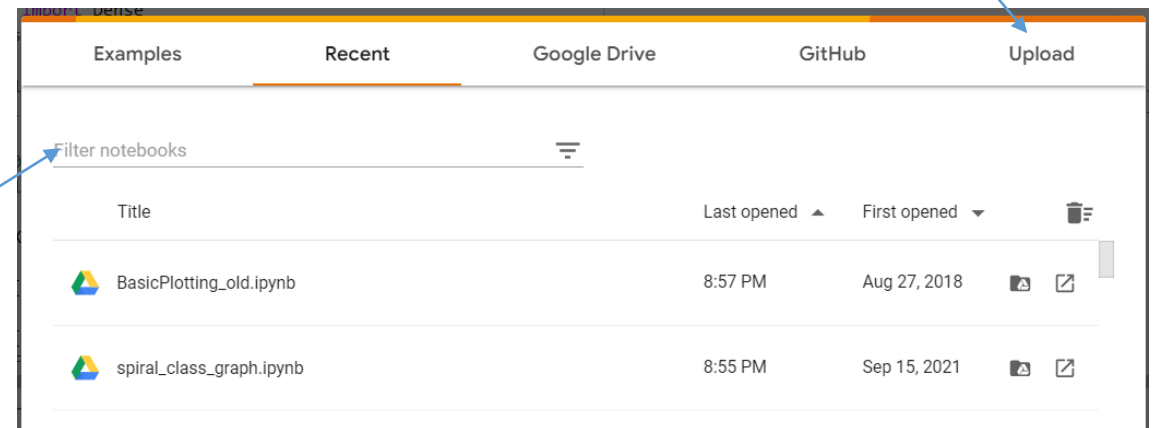


(*) Files are downloaded or created by Anaconda; Cannot open without "Colab app" which is blocked by LTU...

Go to colab.research.google.com using a browser

As long as the ipynb file is under "My Drive/Colab Notebooks" folder, you can find & open it regardless of file type.

You can open ipynb file from your computer. The file will be saved under "My Drive/Colab Notebooks".



Useful Shortcuts in Colab

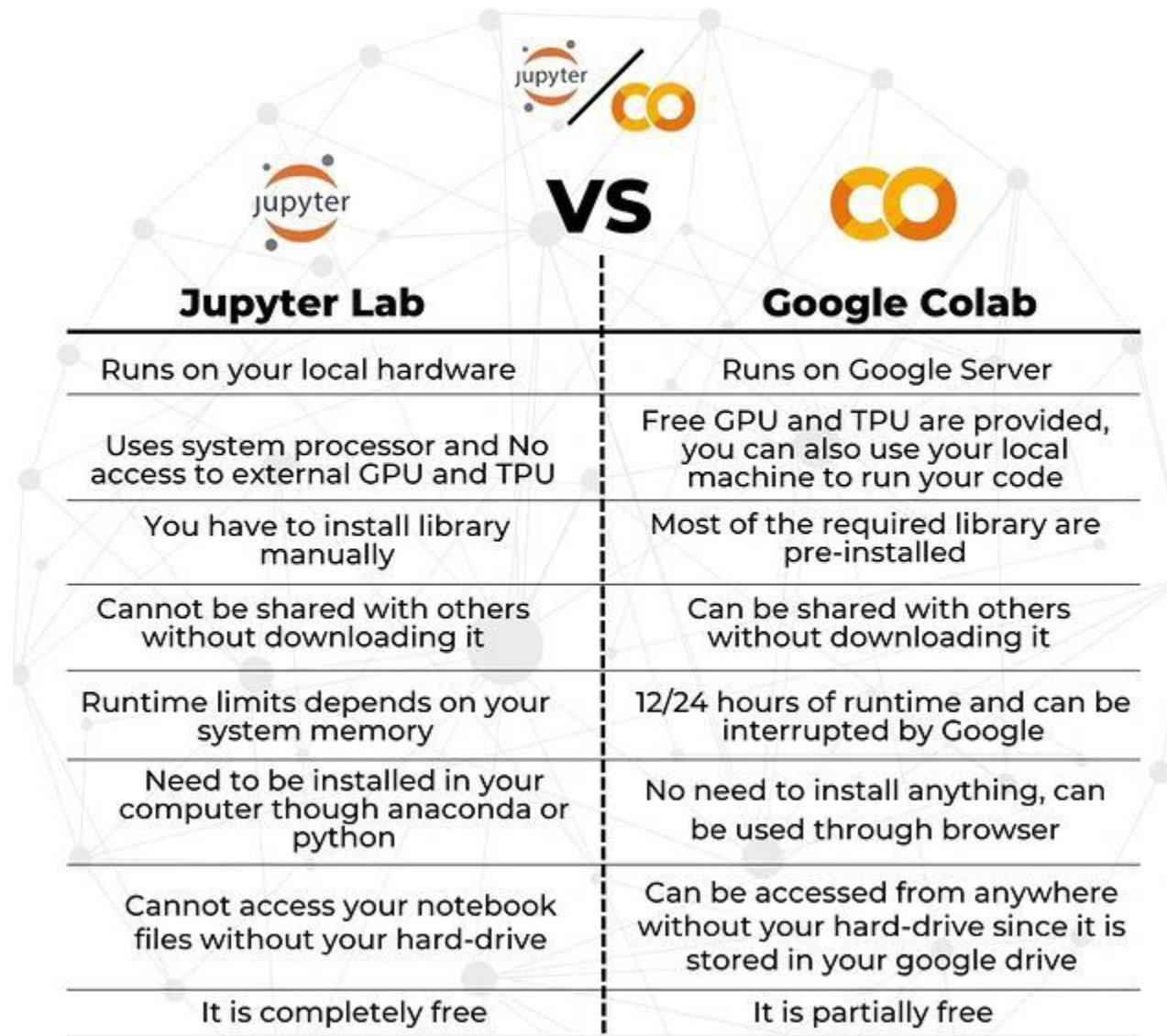
- Ctrl + S -- Save file
- Ctrl + / -- comment/uncomment highlighted section

Markdown Language for Text cells

- <https://en.wikipedia.org/wiki/Markdown>
- <https://www.markdownguide.org/cheat-sheet/>

Deep Learning with Python by François Chollet, 2nd edition

- <https://www.manning.com/books/deep-learning-with-python>
- 3.4.2 Using Colaboratory, pages 73-75



 Jupyter Lab	 Google Colab
Runs on your local hardware	Runs on Google Server
Uses system processor and No access to external GPU and TPU	Free GPU and TPU are provided, you can also use your local machine to run your code
You have to install library manually	Most of the required library are pre-installed
Cannot be shared with others without downloading it	Can be shared with others without downloading it
Runtime limits depends on your system memory	12/24 hours of runtime and can be interrupted by Google
Need to be installed in your computer though anaconda or python	No need to install anything, can be used through browser
Cannot access your notebook files without your hard-drive	Can be accessed from anywhere without your hard-drive since it is stored in your google drive
It is completely free	It is partially free

New: Gemini Generative AI Assistant

